

CSA1717-ARTIFICIAL INTELLIGENCE

PRACTICAL LAB EXPERIMENTS

FORWARD CHAINING

Aim:

To implement the Forward Chaining algorithm in Prolog.

Objective:

To derive new facts from an initial set of facts by repeatedly applying rules.

Algorithm:

- 1. Start.**
- 2. Define initial facts.**
- 3. Define rules.**
- 4. Check whether all conditions of a rule are satisfied.**
- 5. If satisfied, derive the conclusion.**
- 6. Add the new fact to the knowledge base.**
- 7. Repeat until no new facts can be derived.**
- 8. Stop.**

Flowchart:

START

|

v

Load Initial Facts

|

v

Check Rules

|

v

Conditions Satisfied?

/ \

Yes No

| |

Derive Fact Check Next Rule

|

v

Add Fact to KB

|

v

New Fact Generated?

/ \

Yes No

| |

Repeat STOP

Prolog Program:

% Forward Chaining

fact(fever).

fact(cough).

rule([fever, cough], flu).

rule([flu], doctor_visit).

rule([doctor_visit], treatment).

forward :collect_facts(Facts),

forward_chain(Facts, FinalFacts),

write('Final Facts: '),

write(FinalFacts),

nl.

collect_facts(Facts) :findall(Fact,

fact(Fact), Facts).

forward_chain(Facts, FinalFacts) :findall(

Conclusion,

(

rule(Conditions, Conclusion), all_present(Conditions,

Facts),

\+ member(Conclusion, Facts)

),

```

    NewFacts
),
(
    NewFacts = [] ->
    FinalFacts = Facts
;
    append(Facts,      NewFacts,      UpdatedFacts),
    forward_chain(UpdatedFacts, FinalFacts)
).

```

all_present([], _).

```

all_present([H|T],  Facts)
    :member(H,      Facts),
    all_present(T, Facts).

```

Query

?- forward.

Output

Final Facts: [fever,cough,flu,doctor_visit,treatment] true.

Result:

Thus, Forward Chaining is successfully implemented to derive new facts from the knowledge base.